

远程访问

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我们常使用SSH和VNC工具进行远程控制树莓派系统。

1、前期准备

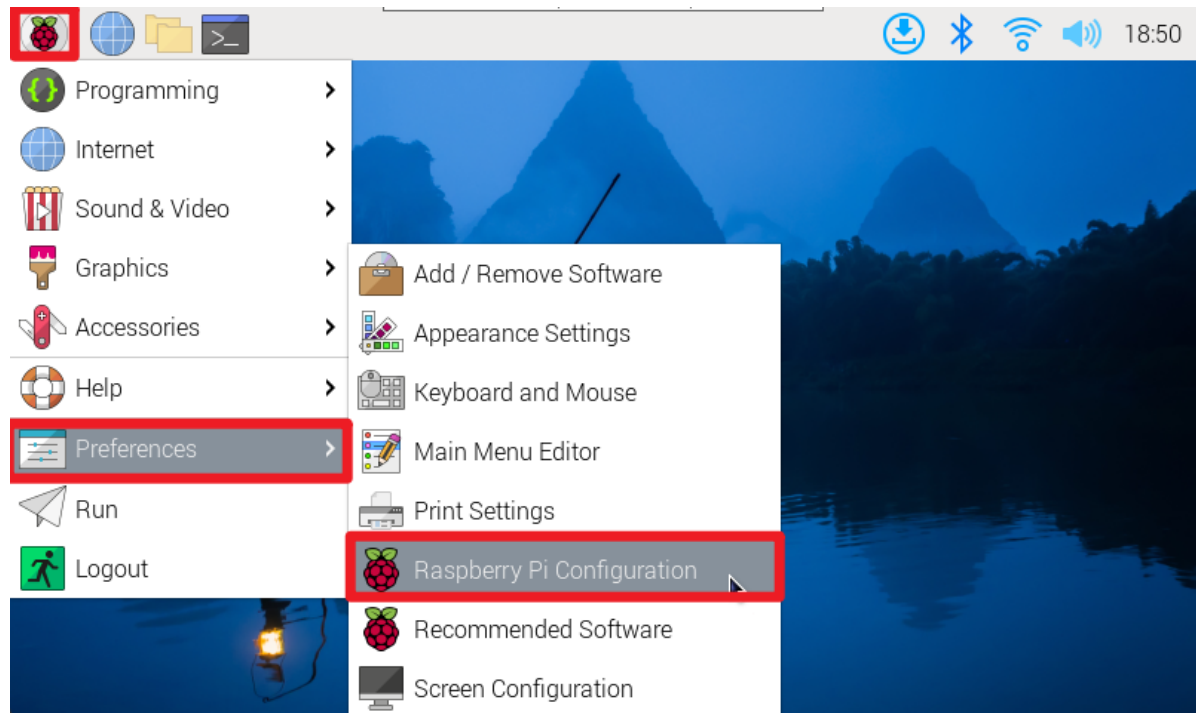
注意：小车镜像已开启SSH和VNC，可以跳过前期准备内容

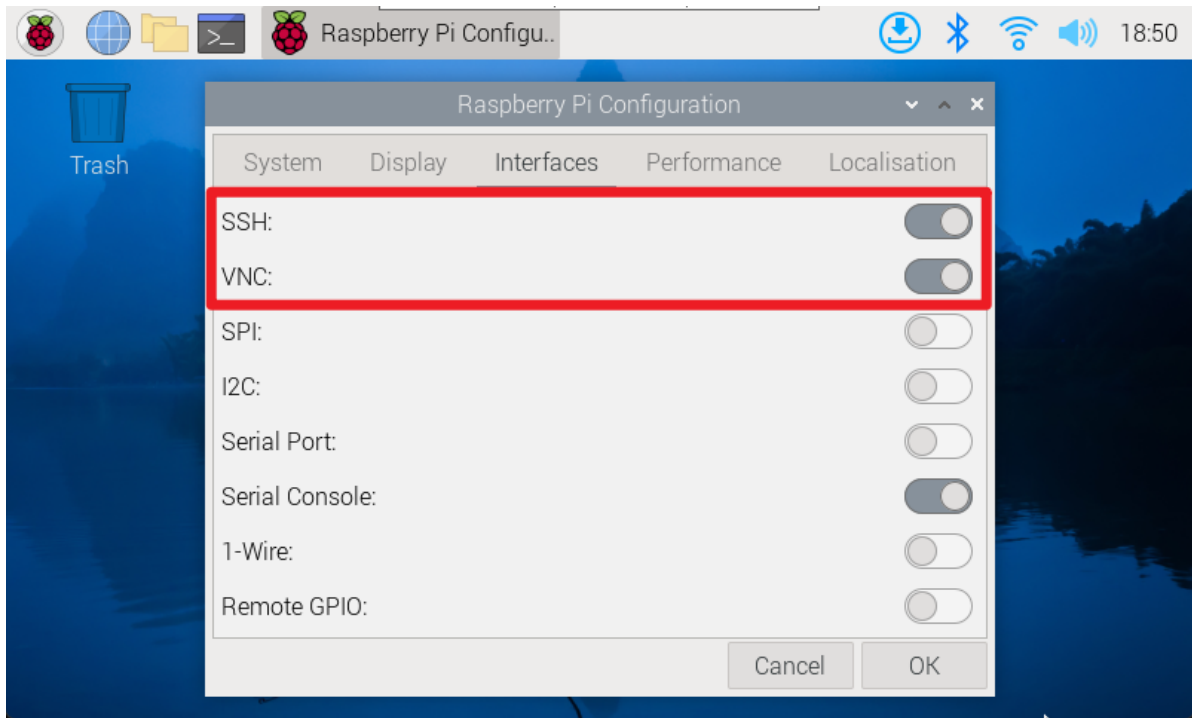
进行SSH、VNC远程登录前，需要在树莓派系统设置或使用raspi-config工具开启SSH和VNC功能。

1.1、开启SSH和VNC

图形化界面

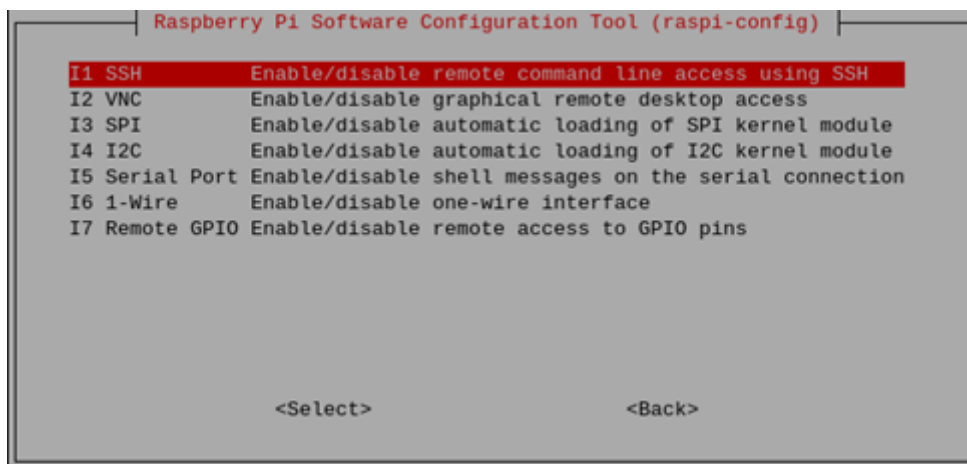
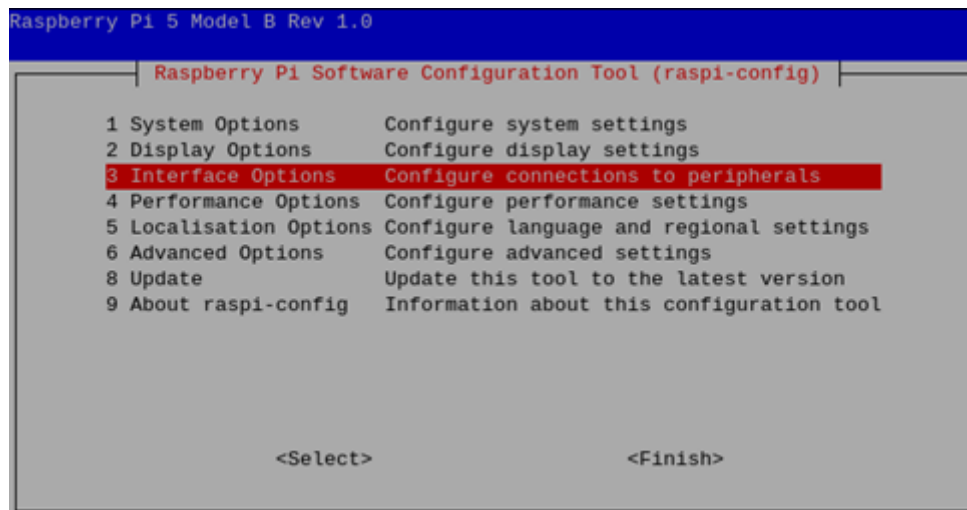
开启SSH和VNC：applications menu → Preferences → Raspberry Pi Configuration

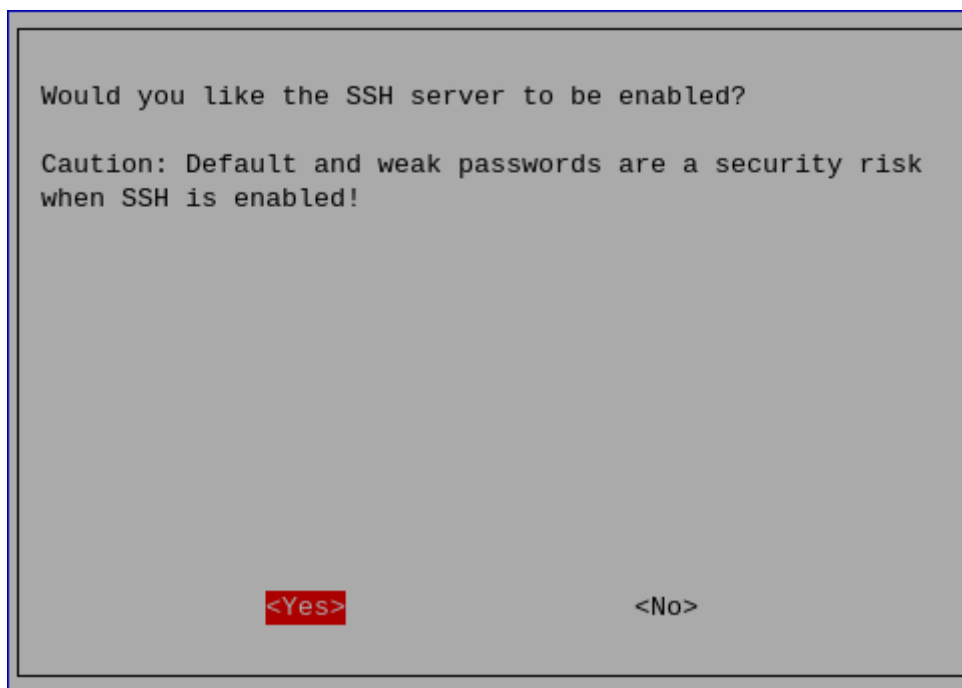




命令行

使用raspi-config工具开启SSH和VNC功能：Interface Options → SSH/VNC: enable





开启VNC功能步骤相同，自行按照上面操作！

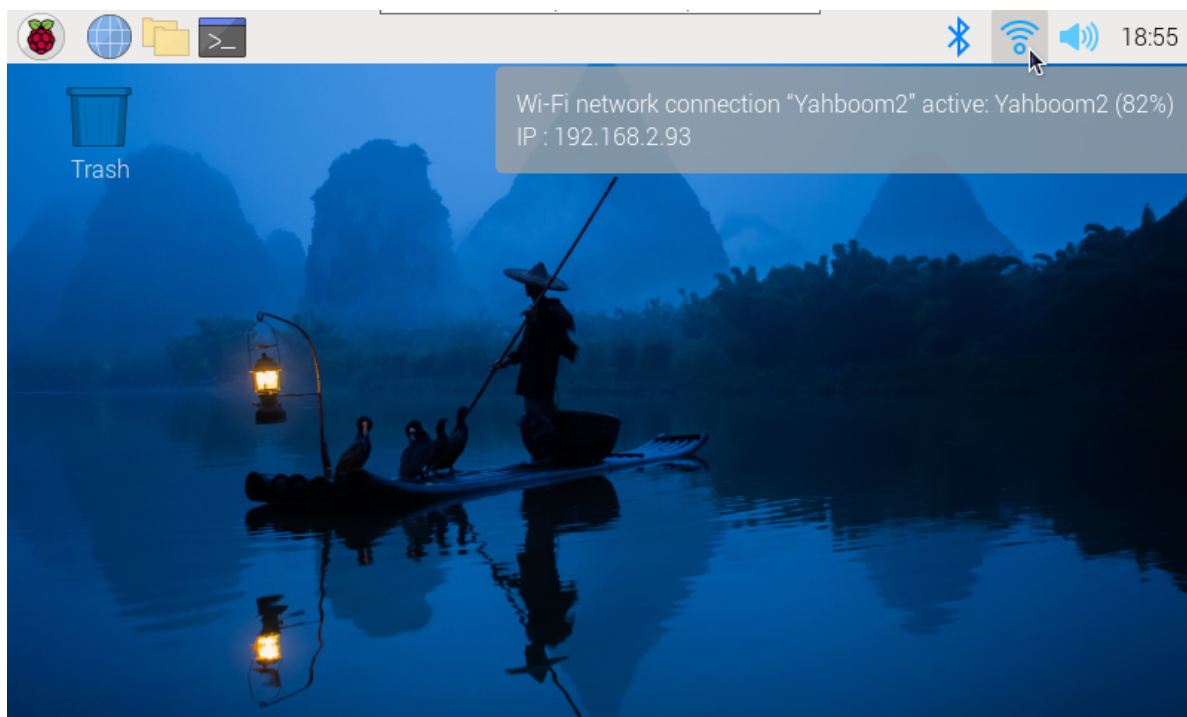
注意：若开启VNC服务失败，检查系统是否有更新；更新软件并重启系统后再重新开启VNC服务。

1.2、获取IP

启用SSH和VNC功能后，就可以根据树莓派的IP进行远程控制！

图像化界面

系统连接WiFi后，将鼠标停留在WiFi图标上，就可以看到对应的IP地址。



使用命令查看IP地址：hostname -I或ifconfig

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ hostname -I  
192.168.2.93  
pi@raspberrypi:~$ ifconfig  
eth0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether d8:3a:dd:bf:89:fd txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
    device interrupt 107  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 182 bytes 15852 (15.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 182 bytes 15852 (15.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.2.93 netmask 255.255.255.0 broadcast 192.168.2.255  
    ether 5c:00:1d:10:00:00 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2、SSH远程控制

获取树莓派主板IP地址后，可以在终端根据树莓派系统的用户名和密码进行SSH远程登录。

以连接小车热点为例，密码：12345678



SSH远程登录命令：ssh 用户名@IP地址

小车的用户名是pi，密码是yahboom,以热点IP地址是192.168.1.11

```
ssh pi@192.168.1.11
```

```
pi@yahboom: ~
Windows PowerShell
版权所有 (C) Microsoft Corporation。保留所有权利。

尝试新的跨平台 PowerShell https://aka.ms/pscore6

PS C:\Users\YB-101> ssh pi@192.168.1.11
The authenticity of host '192.168.1.11 (192.168.1.11)' can't be established.
ECDSA key fingerprint is SHA256:f29Eu99+sFwU8+hG7Mzs0jxoih75pMH180XYsd9pav0.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.11' (ECDSA) to the list of known hosts.
pi@192.168.1.11's password:
Linux yahboom 6.6.31+rpt-rpi-2712 #1 SMP PREEMPT Debian 1:6.6.31-1+rpt1 (2024-05-29) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Aug 14 18:30:39 2024
pi@yahboom:~$
```

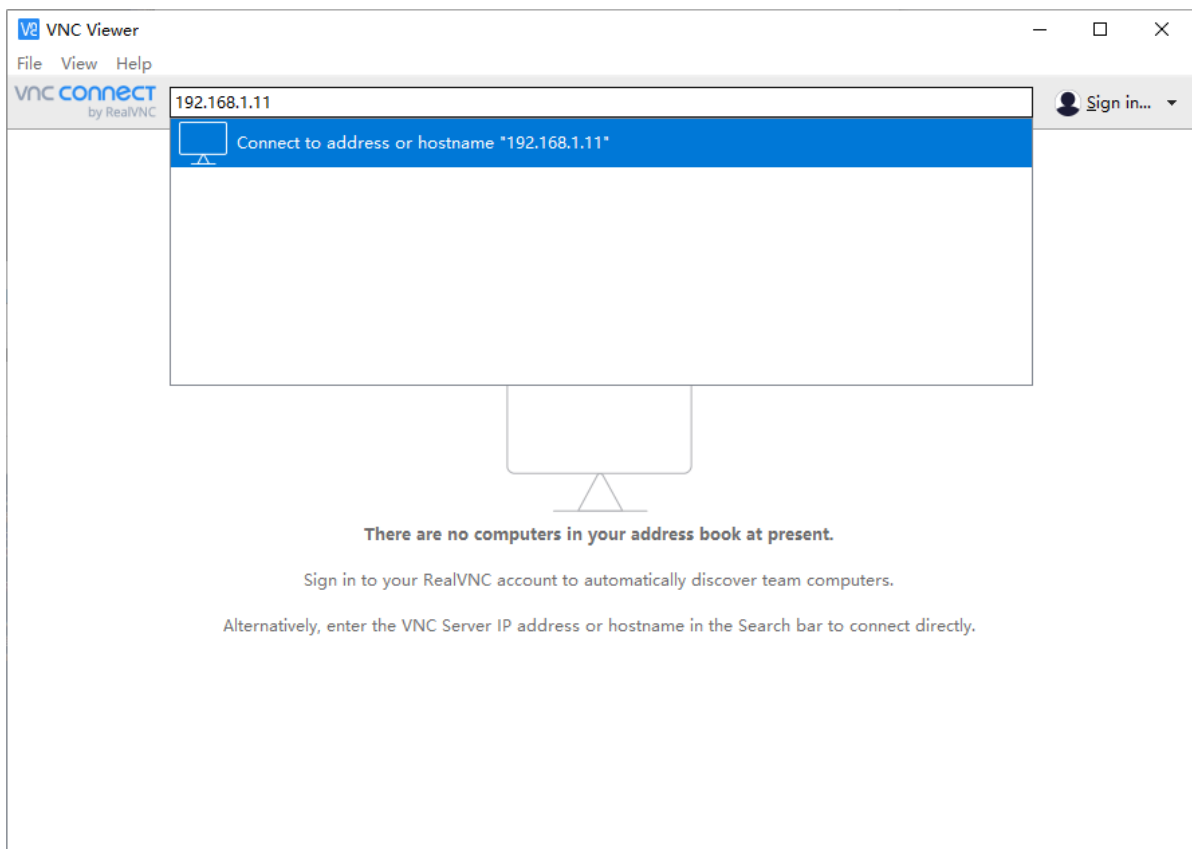
3、VNC远程登录

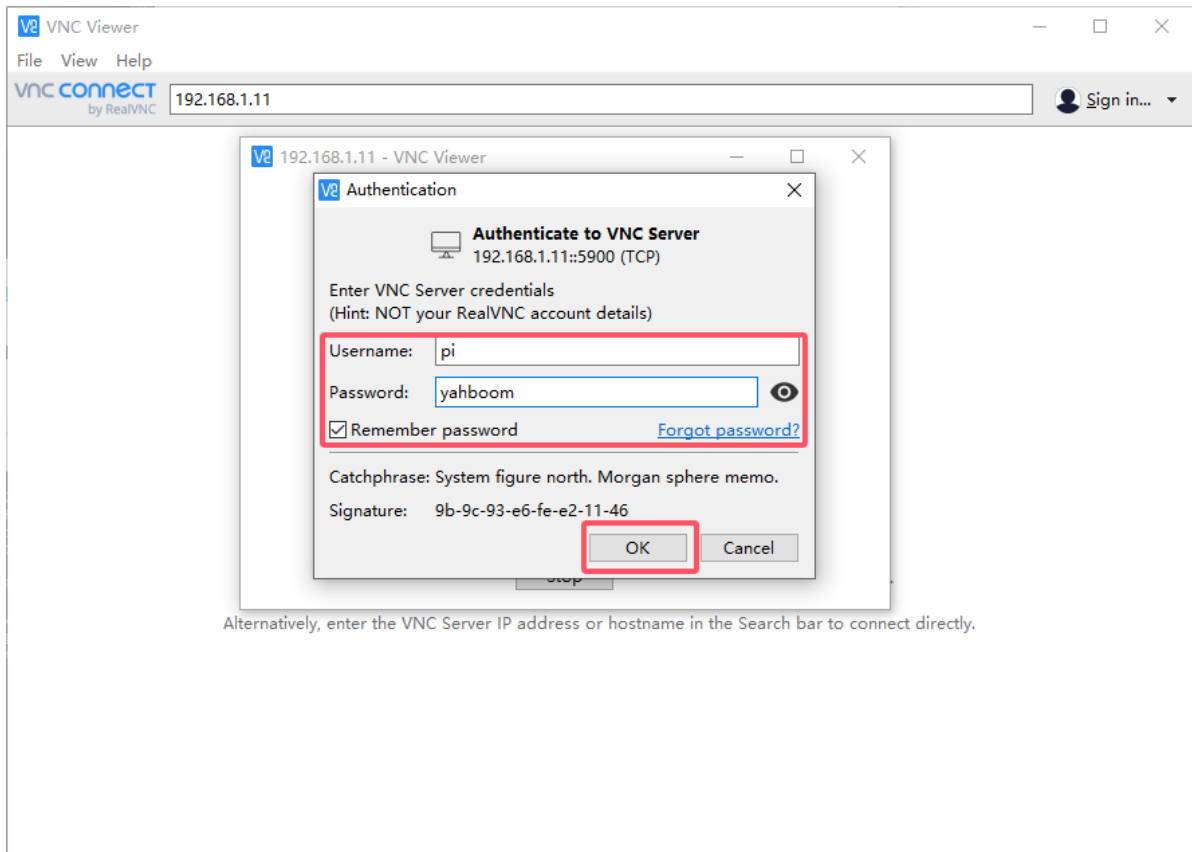
获取树莓派主板IP地址后，可以使用RealVNC Viewer软件进行远程登录。

以连接小车热点为例，密码：12345678



小车登录的用户名是pi，密码是yahboom,热点的IP地址是192.168.1.11





远程登录成功后会显示树莓派系统桌面！

